

Assignment - 9 (Advanced Recursion)

Instructions: Follow the same pattern as before —

- 1. Dry run each input on paper,**
- 2. Write and test your code,**
- 3. Submit both your dry run and final working code.**

1. Count Subsequences with Even Sum

Description: Given an array of integers, count all subsequences whose sum is even.

Input: nums: List[int]

Output: int - Number of subsequences with even sum

Constraints: $1 \leq \text{len}(\text{nums}) \leq 15$

$-100 \leq \text{nums}[i] \leq 100$

Example:

Input: nums = [2, 4, 1]

Output: 4

Explanation: [2], [4], [2, 4], [1, 2, 4] -> all have even sum.

2. Longest Increasing Subsequence (Recursive)

Description: Return the length of the longest increasing subsequence using recursion only.

Input: nums: List[int]

Output: int

Constraints: $1 \leq \text{len}(\text{nums}) \leq 15$

Example:

Input: nums = [10, 9, 2, 5, 3, 7, 101, 18]

Output: 4

3. Subsequences with Alternating Signs

Description: Print all subsequences of the given array where the signs alternate (+/-).

Input: nums: List[int]

Output: List[List[int]]

Constraints: $1 \leq \text{len}(\text{nums}) \leq 10$

No zero in input array.

Example:

Input: nums = [1, -2, 3, -4]

Output: [[1, -2], [-2, 3], [3, -4], [1, -2, 3], [-2, 3, -4], [1, -2, 3, -4]]

4. Subset with Maximum Product

Description: Return the subset (not empty) of nums with the maximum product.

Input: nums: List[int]

Output: List[int]

Constraints: $1 \leq \text{len}(\text{nums}) \leq 10$

Example:

Input: nums = [1, -2, -3, 4]

Output: [-2, -3, 4]

5. Subsets with Given Average

Description: Print all subsets whose average equals k.

Input: nums: List[int], k: float

Output: List[List[int]]

Constraints: $1 \leq \text{len}(\text{nums}) \leq 10$

$0 \leq \text{nums}[i] \leq 50$

Example:

Input: nums = [1, 3, 2, 6], k = 3.0

Output: [[3], [1, 2, 6]]

6. Next Permutation (Recursive)

Description: Given a number in string format, return the next lexicographic permutation.

Input: s: str

Output: str

Constraints: $1 \leq \text{len}(s) \leq 8$

Example:

Input: s = "123"

Output: "132"

7. Word Break with Permutations

Description: Return all permutations of valid word breaks using a given dictionary.

Input: s: str, wordDict: List[str]

Output: List[str]

Constraints: $1 \leq \text{len}(s) \leq 20$

Example:

Input: s = "catsanddog", wordDict = ["cat", "cats", "and", "sand", "dog"]

Output: ["cats and dog", "cat sand dog"]

8. Derangement (No Fixed Position)

Description: Count derangements of n people - permutations where no one is in their original position.

Input: n: int

Output: int

Constraints: $1 \leq n \leq 8$

Example:

Input: $n = 3$

Output: 2

9. N-Sum Problem (Generalized K-Sum)

Description: Find all unique combinations of n numbers that sum up to target.

Input: `nums: List[int], n: int, target: int`

Output: `List[List[int]]`

Constraints: $1 \leq \text{len}(\text{nums}) \leq 20$

Example:

Input: `nums = [1, 2, 3, 4, 5], n = 3, target = 6`

Output: `[[1, 2, 3]]`

10. Partition into K Equal Sum Subsets

Description: Can the array be divided into k subsets with equal sum?

Input: `nums: List[int], k: int`

Output: `bool`

Constraints: $1 \leq \text{len}(\text{nums}) \leq 16$

Example:

Input: `nums = [4, 3, 2, 3, 5, 2, 1], k = 4`

Output: `true`

11. Generate Balanced Parentheses

Description: Generate all combinations of well-formed n pairs of parentheses.

Input: `n: int`

Output: List[str]

Constraints: $1 \leq n \leq 10$

Example:

Input: $n = 3$

Output: ["((()))", "(()())", "()(())", "())()", "())()"]

12. Unique Paths in Grid with Obstacles

Description: Given a grid with obstacles (1), count unique paths from top-left to bottom-right.

Input: grid: List[List[int]]

Output: int

Constraints: $1 \leq \text{rows}, \text{cols} \leq 10$

Example:

Input: grid = [[0,0,0],[0,1,0],[0,0,0]]

Output: 2

13. Binary Strings Without Consecutive 1s

Description: Generate all binary strings of length n with no consecutive 1s.

Input: n : int

Output: List[str]

Constraints: $1 \leq n \leq 10$

Example:

Input: $n = 3$

Output: ["000", "001", "010", "100", "101"]

14. Tug of War Partition

Description: Split the array into two groups of equal size such that absolute difference of their sums is

minimized.

Input: nums: List[int]

Output: int

Constraints: $2 \leq \text{len}(\text{nums}) \leq 20$ and is even

Example:

Input: nums = [1, 2, 3, 4]

Output: 0

15. Palindrome Partitioning

Description: Given a string, return all possible palindrome partitionings.

Input: s: str

Output: List[List[str]]

Constraints: $1 \leq \text{len}(s) \leq 15$

Example:

Input: s = "aab"

Output: [["a","a","b"], ["aa","b"]]